

Advanced Calculus Solution

Unit 4: Integration Techniques

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Problem & Initial Analysis

Definite Integral

Evaluate the following definite integral using substitution and partial fractions if necessary.

$$\int_0^{\pi} \frac{\sin(x)}{1 + \cos^2(x)} dx$$

Step-by-Step Solution

STEP 1**Substitution**

Let $u = \cos(x)$. Then $du = -\sin(x)dx$. Changing the limits: when $x=0$, $u=1$; when $x=\pi$, $u=-1$.

$$du = -\sin(x)dx$$

$$x = 0 \implies u = 1$$

$$x = \pi \implies u = -1$$

STEP 2**Transform Integral**

Substitute u and du into the integral. Don't forget to flip the limits due to the negative sign.

$$-\int_1^{-1} \frac{1}{1+u^2} du = \int_{-1}^1 \frac{1}{1+u^2} du$$

STEP 3

Integration

The integral of $1/(1+u^2)$ is $\arctan(u)$.

$$[\arctan(u)]_{-1}^1$$

Final Result

Evaluating the antiderivative at the limits gives the final answer.

$$\arctan(1) - \arctan(-1) = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$$

Verification

You can verify this by checking the symmetry of the integrand.